

Universal enveloping algebras of finite-dimensional solvable Lie algebras

GPT-6.1-Sol

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1 Introduction

Theorem 1.1. *Let k be a field of characteristic zero. Let L and M be finite-dimensional Lie algebras over k , and suppose that L is solvable. If there is an isomorphism of unital associative k -algebras*

$$\Phi : U_k(L) \longrightarrow U_k(M),$$

then L and M are isomorphic as Lie algebras over k .

We identify a Lie algebra with its image in its enveloping algebra, using the Poincaré–Birkhoff–Witt theorem. For a Lie ideal $J \subseteq g$, the notation $U(g)J$ means the two-sided ideal it generates. Thus $U(g)/U(g)J = U(g/J)$. Write $N(g)$ for the nilradical, and use the convention

$$\gamma_1 N = N, \quad \gamma_{r+1} N = [N, \gamma_r N].$$

The proof separates the information supplied by the associative isomorphism from the deformation argument needed to turn it into a Lie isomorphism. First, we prove that M is solvable and normalize Φ at the augmentation. Locally nilpotent inner derivations then identify the ideal $U(L)[N, N]$ intrinsically. In its metabelian quotient, locally finite inner derivations identify both the polynomial algebra on $N/[N, N]$ and the quotient L/N . Taking the linear part at the augmentation gives an actual graded Lie isomorphism

$$\varphi_0 : \mathrm{gr}_F L \longrightarrow \mathrm{gr}_F M, \quad F^0 L = L, \quad F^r L = \gamma_r N(L) \quad (r \geq 1),$$

whose enveloping map is the associated graded of the normalized Φ .

Next, the corresponding Rees Lie algebras are two polynomial families with the same graded special fibre. Their enveloping algebras are isomorphic by a map equal to the identity at that fibre. After a separate homogenization parameter h is introduced, this map has a uniform Laurent bound on each coefficient of the Rees parameter t . This precise bound allows its use as a gauge transformation in the full Hochschild complex. Affine formality reflects the resulting equivalence to the linear Poisson structures of the two Lie families.

Finally, the first derivative at the origin of the Poisson equivalence produces a Lie isomorphism over a Laurent-series field. Its graded part is a regular arc specializing to φ_0 . The image of the filtered isomorphism scheme on the associated graded is a closed coset. Therefore the fibre with mark φ_0 is nonempty. Its automorphism group is unipotent, and an explicit finite Galois averaging argument descends a marked isomorphism to the original coefficient field.

Only finitely many coefficients of the Lie brackets and of Φ and Φ^{-1} are used to define the data. We can consequently carry out the complex formality argument after embedding a finitely generated characteristic-zero coefficient field into $\mathbb{C}$, and descend the result before extending back to k .

The order of the two formal operations is essential: one first completes in h , then inverts h , and only afterwards completes in t . The relevant completed coefficient spaces have the form $X[[h]][h^{-1}][[t]]$. No lower bound in h uniform in the outer t -degree is required.

2 Intrinsic information in the enveloping algebra

Throughout this section, k has characteristic zero and all Lie algebras are finite dimensional. We use PBW to regard a Lie algebra as a subspace of its enveloping algebra. If J is a Lie ideal of g , then $U(g)J$ denotes the two-sided ideal it generates, and

$$U(g)/U(g)J \simeq U(g/J).$$

The standard Lie-theoretic ingredients are PBW, Lie's theorem, Engel's theorem, and the existence of the nilradical; see [4, 1].

2.1 Solvability, augmentation, and scalar extension

Lemma 2.1. *If L is solvable and $U(L) \simeq U(M)$ as unital k -algebras, then M is solvable.*

Proof. Over an algebraically closed field, a finite-dimensional Lie algebra is solvable if and only if all its finite-dimensional irreducible modules are one dimensional. The forward implication is Lie's theorem. For the converse, a composition series of the adjoint module becomes a full invariant flag. The adjoint image is therefore upper triangular and solvable; its central kernel implies solvability of the original algebra.

For an algebraic closure Ω/k , PBW gives

$$\Omega \otimes_k U(g) \simeq U(\Omega \otimes_k g).$$

An associative algebra isomorphism transports simple modules without changing their underlying vector spaces. Thus the preceding criterion passes from L_Ω to M_Ω . Derived series commute with field extension, and faithful scalar extension reflects vanishing, so solvability descends to k . $\square$

Denote the initially given associative isomorphism by Φ_{raw} .

Lemma 2.2. *An isomorphism $\Phi_{\text{raw}} : U(L) \rightarrow U(M)$ can be made augmentation preserving by composing it with an explicit automorphism of its source.*

Proof. Let ε_g denote the standard augmentation and set $\chi = \varepsilon_M \Phi_{\text{raw}}|_L$. Then $\chi([x, y]) = 0$. For any Lie character λ , the assignment $x \mapsto x + \lambda(x)1$ preserves the enveloping relations and extends to an automorphism τ_λ , with inverse $\tau_{-\lambda}$. Consequently

$$\Phi = \Phi_{\text{raw}} \circ \tau_{-\chi}, \quad \varepsilon_M \Phi = \varepsilon_L.$$

It suffices to check the latter identity on the Lie generators. $\square$

From now on, Φ denotes this specified normalized isomorphism; Φ_{raw} denotes the initially given one.

Write $N(g)$ for the nilradical and $\gamma_1 N = N$, $\gamma_{r+1} N = [N, \gamma_r N]$ for its lower central series. We will need compatibility with arbitrary field extensions, including transcendental ones. A finite system of trace equations provides this directly. The use of the full associative algebra generated by the adjoint operators is important: testing only the individual adjoint operators would replace these equations by the Killing form, which does not give the assertion below.

Lemma 2.3. *Let $\mathcal{A} \subseteq \text{End}_k(g)$ be the unital associative algebra generated by $\text{ad}(g)$. Then*

$$N(g) = \{x \in g : \text{tr}(\text{ad}(x)T) = 0 \text{ for every } T \in \mathcal{A}\}.$$

Proof. Call the displayed right-hand side J . Cyclicity of trace gives

$$\text{tr}(\text{ad}([y, x])T) = \text{tr}(\text{ad}(x)(T \text{ad}(y) - \text{ad}(y)T)),$$

so J is an ideal. If $x \in J$, choosing $T = (\text{ad } x)^{r-1}$ shows that all positive power traces of $\text{ad } x$ vanish. In dimension n , the first n Newton identities, whose integer denominators are invertible in k , make all the nonleading coefficients of its characteristic polynomial zero. Cayley–Hamilton then implies that $\text{ad } x$ is nilpotent. Its restriction to J is nilpotent, so Engel’s theorem makes J nilpotent. Hence $J \subseteq N(g)$.

Conversely, let I be a nilpotent ideal and $x \in I$. Every element of $\mathcal{A}$ preserves every Lie ideal, in particular the ideals $\gamma_r I$. Thus $\text{ad}(x)T$ sends g into I and sends $\gamma_r I$ into $\gamma_{r+1} I$. It is nilpotent and has trace zero. Therefore $I \subseteq J$, proving the reverse inclusion. $\square$

Corollary 2.4. *For every field extension K/k ,*

$$N(g_K) = N(g)_K, \quad \gamma_r N(g_K) = (\gamma_r N(g))_K.$$

Proof. Choose a basis $T_1, \dots, T_s$ of $\mathcal{A}$. The nilradical is the kernel of the linear map $x \mapsto (\text{tr}(\text{ad}(x)T_i))_i$. Each extended T_i belongs to the adjoint envelope over K . By [theorem 2.3](#), the new nilradical satisfies these extended equations. Kernels commute with field extension, giving $N(g_K) \subseteq N(g)_K$. The reverse inclusion follows because extension preserves nilpotent ideals. The lower central identities follow by extending the successive images of the bracket maps. $\square$

2.2 Weights and two derivation lemmas

Lemma 2.5. *For a solvable g , put $N = N(g)$, $Q = g/N$, and $V = N/[N, N]$. Then Q is abelian. Over an algebraic closure, the simultaneous weights of its action on V span Q^* ; each such weight has a nonzero common eigenvector.*

Proof. Lie’s theorem makes $\text{ad}(g)$ upper triangular over an algebraic closure. Its commutators are strictly upper triangular, so Engel’s theorem shows that $[g, g]$ is nilpotent. This descends to k , giving $[g, g] \subseteq N$. The action on V factors through the abelian quotient Q .

We first observe that a derivation D of a nilpotent algebra N is nilpotent if its action on V is nilpotent. Iterated brackets give surjective D -equivariant maps

$$V^{\otimes r} \longrightarrow \gamma_r N / \gamma_{r+1} N.$$

Replacing one argument by an element of $[N, N]$ changes the bracket by an element of $\gamma_{r+1} N$, which justifies these maps. If $D^s|_V = 0$, its action on $V^{\otimes r}$ is a sum of r commuting operators and has power $r(s-1)+1$ equal to zero. Thus D is nilpotent on every successive quotient and hence on N .

Work now over the algebraic closure, using [theorem 2.4](#). Let Q_0 be the common kernel of all weights and H its inverse image in g . This is an ideal. Every $x \in H$ acts nilpotently on V , therefore on N , and acts trivially on H/N . Hence every $\text{ad}_H x$ is nilpotent. Engel’s theorem and maximality of N imply $H = N$, so $Q_0 = 0$, proving separation. On each simultaneous generalized weight space, the commuting nilpotent parts have a nonzero common kernel. This supplies a genuine common eigenvector for each weight. $\square$

The proof also covers $Q = 0$, when separation in the zero dual space is automatic. In general, a finite subfamily of the weights already separates Q , since Q is finite dimensional. Zero weights and nonsemisimple actions are allowed. Separation of the weights, rather than mere faithfulness of the action, is essential below.

Lemma 2.6. *Suppose A has an exhaustive increasing algebra filtration $F_d A$, and a locally finite derivation D satisfies $D(F_d A) \subseteq F_{d+r} A$ for a fixed $r > 0$. Its induced homogeneous derivation of degree r on $\text{gr}_F A$ is locally nilpotent.*

Proof. Lift a homogeneous symbol of degree d to $b \in F_d A$. The finite-dimensional orbit span of b lies in some $F_B A$. The j th iterate of the induced operator is the symbol of $D^j b$ in degree $d + jr$, hence is zero once $d + jr > B$. Finite sums of homogeneous symbols are handled by taking a maximum. The individual $F_d A$ need not be finite dimensional. $\square$

Lemma 2.7. *If D is a locally nilpotent derivation of a commutative domain of characteristic zero and $0 \neq u$ satisfies $D(u) \in uA$, then $D(u) = 0$.*

Proof. For $a \neq 0$ set $\nu(a) = \max\{j : D^j a \neq 0\}$. The iterated Leibniz formula gives $\nu(ab) = \nu(a) + \nu(b)$: the unique surviving term at that order has a nonzero binomial coefficient:

$$D^{\nu(a)+\nu(b)}(ab) = \binom{\nu(a)+\nu(b)}{\nu(a)} D^{\nu(a)} a D^{\nu(b)} b.$$

If $D(u) = uv \neq 0$, then $\nu(u) - 1 = \nu(Du) = \nu(u) + \nu(v)$, a contradiction. $\square$

2.3 The abelian-extension calculation

For an associative algebra A , let $\text{LN}(A)$ and $\text{LF}(A)$ be the sets of elements whose inner derivations are respectively locally nilpotent and locally finite. Local finiteness means that each vector has a finite-dimensional span of iterates. These definitions are intrinsic to the associative algebra.

Proposition 2.8. *Let H be an abelian ideal of g with abelian quotient $T = g/H$. Suppose the simultaneous weights of T on H after algebraic closure span T^* . Then, inside $U(g)$,*

$$\text{LN}(U(g)) = S(H), \quad \text{LF}(U(g)) = g + S(H) = s(T) + S(H)$$

for every linear section $s : T \rightarrow g$.

Proof. First work over an algebraically closed field. Choose lifts $t_1, \dots, t_d$ of a basis of T , and put $\delta_i = \text{ad}(t_i)|_{S(H)}$. These are commuting linear polynomial derivations: $[\delta_i, \delta_j] = \text{ad}([t_i, t_j])|_{S(H)} = 0$. We retain the possibly nonzero cocycle $c_{ij} = [t_i, t_j] \in H$.

Filter by the number of quotient letters: H has degree zero and each t_i degree one, and we set $F_{-1} = 0$. PBW gives

$$\text{gr } U(g) = S(H)[\xi_1, \dots, \xi_d].$$

Indeed $[t_i, H] \subseteq H$ lowers degree by one and $[t_i, t_j] \in H$ lowers degree by two. In particular $[F_m, F_n] \subseteq F_{m+n-1}$.

Let $a \in \text{LF}(U(g))$ have degree $m \geq 2$ and leading symbol p . The derivation $\text{ad } a$ induces a derivation D of positive degree $m - 1$, locally nilpotent by [theorem 2.6](#). For a common eigenvector $0 \neq w \in H$ of weight λ , the commutator Leibniz rule yields

$$D(w) = w \sum_i \lambda(t_i) \frac{\partial p}{\partial \xi_i}.$$

Here each occurrence of t_i contributes $\lambda(t_i)$; further reorderings lower degree. By [theorem 2.7](#), the sum is zero. Separation by the weights forces every partial derivative of p to vanish. Characteristic zero makes p independent of the ξ_i , contradicting its positive homogeneous degree. Thus $m \leq 1$.

The cocycle has not been set to zero: its contributions to $[a, t_j]$ have degree at most $m - 1$, whereas the symbol defining $D(t_j)$ is taken in degree m . Terms from commuting coefficients in $S(H)$ past t_j may survive in degree m and are retained. This verifies the degree estimate also on the quotient generators.

Write, by PBW,

$$a = \sum_i a_i(H) t_i + b(H).$$

On the invariant subalgebra $S(H)$, its inner derivation is the locally finite derivation $\sum_i a_i \delta_i$. If the largest ordinary polynomial degree of the coefficients a_i is $e > 0$, its leading derivation is $E = \sum_i (a_i)_e \delta_i$, of positive degree e . Again it is locally nilpotent. For each eigenvector as above,

$$E(w) = w \sum_i \lambda(t_i) (a_i)_e.$$

Divisibility and separation force all $(a_i)_e$ to vanish, a contradiction. Thus every a_i is constant, proving the required inclusion for LF.

Conversely take $a = s(t) + b(H)$ and give the basis elements of H weight one and the lifts weight $e = \max(1, \deg b)$, using $e = 1$ when $b = 0$. Bounded weighted PBW spaces are finite dimensional. The relations have nonincreasing weight, and

$$[a, H] \subseteq H, \quad [a, s(q)] = [s(t), s(q)] - \delta_q b$$

has polynomial degree at most e . Thus $\text{ad } a$ preserves every such space and is locally finite. This proves the LF equality with the cocycle included.

For $b \in S(H)$, $\text{ad } b$ vanishes on $S(H)$ and strictly lowers the number of quotient letters; it is locally nilpotent. Conversely $\text{LN} \subseteq \text{LF}$, so an element of LN is $s(t) + b(H)$. If $t \neq 0$, some weight has $\lambda(t) \neq 0$, and on its eigenvector

$$(\text{ad}(s(t) + b))^r w = \lambda(t)^r w \neq 0$$

for every r . Hence $t = 0$.

Finally, the two local properties of an operator defined over k can be checked after field extension. In the forward direction each extended vector is a finite sum of pure tensors: take the maximum of finitely many nilpotence exponents, or the sum of finitely many finite-dimensional orbit spaces. Conversely, an infinite independent family of iterates remains independent after extension, and a nonzero iterate remains nonzero. These assertions concern an operator defined over k and do not require any closure property of the sets LN and LF in an arbitrary associative algebra. The memberships in $S(H)$ and $s(T) + S(H)$ are vanishing conditions on PBW coefficients and therefore descend. This proves the assertion over the original field. The equality $g + S(H) = s(T) + S(H)$ follows from $g = H + s(T)$. $\square$

Remark 2.9. The extension in [theorem 2.8](#) need not split. For example, let g have basis t_1, t_2, w_1, w_2, z and only the nonzero brackets

$$[t_1, w_1] = w_1, \quad [t_2, w_2] = w_2, \quad [t_1, t_2] = z.$$

Then $H = \langle w_1, w_2, z \rangle$ is abelian and its two nonzero weights separate g/H . Changing the lifts adds only w_1, w_2 components to their bracket, so its z component cannot be removed. The proposition applies to this extension without change.

Proposition 2.10. *For a solvable g with nilradical N ,*

$$U(g)[N, N] = \langle [a, b] : a, b \in \text{LN}(U(g)) \rangle_{\text{ideal}}.$$

Consequently every associative enveloping-algebra isomorphism preserves this ideal, without an augmentation hypothesis.

Proof. For $x \in N$, $(\text{ad } x)^r(g) \subseteq \gamma_r N$ for $r \geq 1$. Hence $\text{ad } x$ is nilpotent on the Lie generators and locally nilpotent on their products, by the iterated Leibniz rule. Thus $N \subseteq \text{LN}(U(g))$, giving one inclusion.

In the reverse direction, set $\bar{g} = g/[N, N]$. Its abelian ideal $V = N/[N, N]$ and quotient $Q = g/N$ satisfy [theorem 2.5](#). By [theorem 2.8](#), $\text{LN}(U(\bar{g})) = S(V)$ is commutative. Locally nilpotent inner derivations descend through associative quotients, so all the displayed commutators vanish in $U(\bar{g})$. Finally, an algebra isomorphism preserves LN and commutators. $\square$

We have therefore recovered an actual quotient enveloping algebra and, inside it, an actual polynomial subalgebra through intrinsic associative conditions. For two isomorphic enveloping algebras of solvable Lie algebras, the induced quotient isomorphism carries $S(N/[N, N])$ onto the corresponding polynomial algebra on the other side, and preserves the set of linear Lie elements plus that polynomial algebra. These are the inputs for recovering the filtration in the next section; preservation of the original subalgebra $U(N)$ has not been asserted.

3 The intrinsic filtration and its graded map

Let k have characteristic zero, let L, M be finite-dimensional, and suppose that L is solvable and that $\Phi_{\text{raw}} : U(L) \xrightarrow{\sim} U(M)$ is a k -algebra isomorphism. The preceding results imply that M is solvable. Write

$$N = N(L), \quad N' = N(M), \quad Q = L/N, \quad Q' = M/N', \quad I = U(L)N, \quad I' = U(M)N'.$$

All ideals generated in an enveloping algebra are two-sided. We identify Lie elements with their canonical images by PBW.

We first fix the normalization from [theorem 2.2](#). If $\chi(x) = \varepsilon_M(\Phi_{\text{raw}}(x))$, then

$$\Phi = \Phi_{\text{raw}} \circ \tau_{-\chi}, \quad \tau_{-\chi}(x) = x - \chi(x)1$$

preserves augmentation. Both Φ and Φ^{-1} will always refer to these specified maps. This is essential: the automorphism $x \mapsto x + 1$ of $k[x]$ does not preserve (x) . There is no claim that the unnormalized Φ_{raw} preserves I .

3.1 The metabelian map and its linear terms

Put $E = U(L)[N, N]$, and define E' similarly. By [theorem 2.10](#), these ideals are intrinsically generated by commutators of locally nilpotent inner elements. Thus even Φ_{raw} carries E onto E' . In particular, Φ induces an augmentation-preserving isomorphism

$$\bar{\Phi} : U(g) \xrightarrow{\sim} U(g'), \quad g = L/[N, N], \quad g' = M/[N', N'].$$

The abelian ideals $V = N/[N, N]$ and $V' = N'/[N', N']$ embed as polynomial subalgebras $S(V)$ and $S(V')$. Choose linear sections $s : Q \rightarrow g$ and $s' : Q' \rightarrow g'$. They are not assumed to be Lie maps. The intrinsic descriptions in [theorem 2.8](#) give

$$\text{LN}(U(g)) = S(V), \quad \text{LF}(U(g)) = s(Q) + S(V).$$

The analogous equalities hold for g' . Consequently the actual map has the form

$$\bar{\Phi}(v) = F(v), \quad \bar{\Phi}(s(q)) = s'(Tq) + b_q, \quad (3.1)$$

where $F : S(V) \xrightarrow{\sim} S(V')$ is an algebra isomorphism, $T : Q \xrightarrow{\sim} Q'$ is linear, and $q \mapsto b_q \in S(V')$ is linear. Indeed, T is the induced map on LF/LN , and PBW identifies this vector-space quotient with Q . The decomposition is unique since $s'(Q') \cap S(V') = 0$. Augmentation preservation says that $F(v)$ and b_q have zero constant term.

Let $P : V \rightarrow V'$ be the linear term of F at the origin, and let $B(q) \in V'$ be the linear term of b_q . The inverse of F also fixes the origin, so comparison of linear terms in both compositions proves that P is invertible. To keep track of the extension cocycle, write

$$[s(q), v] = A_q v, \quad [s(q), s(r)] = c(q, r) \in V,$$

and use primes for the corresponding target data. Applying $\bar{\Phi}$ to these commutators and taking linear terms gives

$$PA_q = A'_{Tq}P, \quad (3.2)$$

$$Pc(q, r) = c'(Tq, Tr) + A'_{Tq}B(r) - A'_{Tr}B(q). \quad (3.3)$$

Here each A'_{Tq} extends as a derivation preserving ordinary polynomial degree, so higher terms of F or b_q cannot contribute. Thus the linear map $v + s(q) \mapsto Pv + s'(Tq) + B(q)$ is a Lie isomorphism of the metabelian quotients. In particular, the argument retains the possibly nonzero cocycle rather than assuming a split extension.

Proposition 3.1. *The normalized map satisfies $\Phi(I^d) = (I')^d$ for every $d \geq 0$. On $U(L)/I = S(Q)$, its induced map is exactly $S(T)$. Under the natural right-module identifications*

$$I/I^2 \simeq V \otimes_k S(Q), \quad I'/(I')^2 \simeq V' \otimes_k S(Q'),$$

its first graded map is $P \otimes S(T)$.

Proof. The ideal I/E in $U(g)$ is generated by the augmentation ideal $(V) \subset S(V)$. Since F and F^{-1} fix the origin, they carry (V) and (V') onto one another. Lifting through E, E' yields $\Phi(I) = I'$, hence equality of all their powers. In (3.1), the terms b_q vanish modulo I' ; the quotient map is therefore $S(T)$.

The PBW description of the first quotients follows from [theorem 3.2](#) below: order the nilradical variables before the section variables. The monomials of weight one are a single representative of V , followed by an arbitrary monomial in Q . Their classes give the displayed right-module identifications. The relation $qv - vq = A_q v$ describes the left action.

Since $[N', N'] \subset (I')^2$, we have $E' \subset (I')^2$. For $n \in N$, its image on the first layer may therefore be computed from $F(\bar{n})$ in the metabelian quotient. This polynomial contains no Q' -variables. Its terms of degree at least two in V' vanish modulo $(I')^2$, leaving precisely $P(\bar{n})$. Right multiplication and the already identified quotient map prove the asserted formula on the whole module. In particular the actual constant subspace $V \otimes 1$ maps onto $V' \otimes 1$; an equality of module ranks would not give this conclusion.

Explicitly, on this first quotient the left action is

$$q \cdot (v \otimes h) = A_q v \otimes h + v \otimes qh.$$

Its transport is consistent with (3.2). Although $\Phi(s(q)) - s'(Tq)$ may contain terms from I' , commuting any such term with an element of I' gives an element of $(I')^2$. Thus these terms introduce no correction to the first-layer action. This also explains why the term $B(q)$, which is required to retain the metabelian extension cocycle, does not enter the degree-zero map of the descending associated graded algebra. $\square$

3.2 Ideal powers and weighted PBW

Define a descending Lie filtration by

$$F^0 L = L, \quad F^d L = \gamma_d N \quad (d \geq 1), \quad \gamma_1 N = N, \quad \gamma_{d+1} N = [N, \gamma_d N].$$

Choose a basis e_i adapted to this finite flag, with weight $w_i = d$ on a representative of $F^d L / F^{d+1} L$. Thus the complement of N has weight zero. Let W_d be the span of ordered PBW monomials of total weight at least d .

Lemma 3.2. *For every $d \geq 0$, $I^d = W_d$. For $d \geq 1$,*

$$L \cap I^d = \gamma_d N.$$

Proof. The lower central series satisfies

$$[\gamma_i N, \gamma_j N] \subset \gamma_{i+j} N, \quad [L, \gamma_j N] \subset \gamma_j N.$$

Accordingly, every basis vector occurring in $[e_i, e_j]$ has weight at least $w_i + w_j$. In a PBW reordering step $e_j e_i = e_i e_j + [e_j, e_i]$, no resulting term has smaller weight. Repeated reordering shows that $W_d W_e \subset W_{d+e}$. Since $W_0 = U(L)$, W_1 is a two-sided ideal containing N . Hence $I^d \subset W_d$.

Conversely, induction using $[I^r, I^s] \subset I^{r+s}$ gives $\gamma_j N \subset I^j$. Every basis vector e_i belongs to I^{w_i} , including the weight-zero case. A PBW monomial of total weight w therefore lies in I^w , and in I^d when $w \geq d$. This proves the reverse inclusion. Finally, a linear Lie element has only the degree-one PBW coordinates of its basis expansion. Its belonging to W_d is exactly its belonging to the span of the e_i with $w_i \geq d$, which is $\gamma_d N$. $\square$

The use of zero weights is deliberate. Weight pieces need not be finite-dimensional, but every PBW expansion has finite support. In particular $\bigcap_d I^d = 0$, and no finite-dimensionality of the individual graded pieces has been assumed. Indeed, for a nonzero element choose any monomial with nonzero PBW coefficient. Membership in every W_d would require that one fixed monomial to have arbitrarily large weight. The basis and ordering only describe these subspaces: their equality with the intrinsically specified powers I^d makes the resulting filtration independent of both choices.

3.3 The prescribed graded Lie identification

The quotient Q acts canonically on $\text{gr}_\gamma N$. Changing a lift of q by $n \in N$ changes its action on $\gamma_d N$ only in $\gamma_{d+1} N$. The commutator of two lifted actions is inner by an element of N and hence acts trivially on each graded piece. Thus

$$\text{gr}_F L = Q \ltimes \text{gr}_\gamma N,$$

with Q in degree zero. Sending a Lie class to its ideal-adic symbol defines a graded algebra isomorphism

$$U(\text{gr}_F L) \xrightarrow{\sim} \text{gr}_I U(L). \quad (3.4)$$

Indeed, the filtration bracket identities make this a homomorphism, and [theorem 3.2](#) identifies the ordered PBW monomials of each weight as bases on both sides.

Proposition 3.3. *Under (3.4) and its target analogue, the actual map $\text{gr}_I \Phi$ is $U(\varphi_0)$ for a graded Lie isomorphism*

$$\varphi_0 : Q \ltimes \text{gr}_\gamma N \xrightarrow{\sim} Q' \ltimes \text{gr}_\gamma N'.$$

There is a filtered linear isomorphism $f : L \rightarrow M$ inducing φ_0 , such that, for $x \in F^d L$,

$$\Phi(x) - f(x) \in (I')^{d+1}.$$

Its inverse satisfies the reciprocal congruences.

Proof. The Lie algebra $\text{gr}_\gamma N$ is generated by its degree-one subspace V . This follows inductively from $\gamma_{d+1} N = [N, \gamma_d N]$: replacing an input from N by an element of $\gamma_2 N$ changes the bracket only in the next layer. By [theorem 3.1](#), $\text{gr}_I \Phi$ carries V onto V' . Preservation of associative commutators therefore carries their generated Lie subalgebras onto one another. On the degree-zero linear subspace its restriction is T . Together these restrictions give the stated graded Lie isomorphism. The enveloping algebras are generated by these Lie subspaces, so equality on them proves $\text{gr}_I \Phi = U(\varphi_0)$ on the entire graded algebra. In particular, the restriction to Q is the map T extracted from the given metabelian quotient map, and the restriction to V is its actual derivative P . No preservation of a coproduct is part of the hypotheses. What singles out the relevant Lie subspace inside the graded associative algebra is the constant first layer, together with generation by commutators, rather than any additional Hopf-algebra structure imposed on the original isomorphism.

For completeness, the linear lift can be chosen while retaining this exact map. Choose an adapted basis of L , and choose in M a representative of $\varphi_0([e_i])$ in degree w_i . Extend these choices linearly to f . The equality of symbols gives

$$f(F^d L) \subset F^d M, \quad \Phi(x) - f(x) \in (I')^{d+1} \quad (x \in F^d L).$$

Apply the same construction to Φ^{-1} , obtaining g . Ideal-power preservation and [theorem 3.2](#) imply

$$(gf - \text{id})(F^d L) \subset F^{d+1} L, \quad (fg - \text{id})(F^d M) \subset F^{d+1} M.$$

Both error operators are nilpotent because the Lie filtrations terminate. Hence both compositions are invertible, so f is an isomorphism. Its inverse is filtered: for a nonzero vector of highest filtration degree, invertibility of the associated graded map prevents its image from entering the next layer. Applying Φ^{-1} to the forward congruence now proves the reciprocal one for f^{-1} . $\square$

In particular, with $e'_i = f(e_i)$, the two adapted bases have the same weights and satisfy

$$\Phi(e_i) - e'_i \in (I')^{w_i+1}, \quad \Phi^{-1}(e'_i) - e_i \in I^{w_i+1}. \quad (3.5)$$

These are the fixed leading data for the Rees construction in the next section. Its zero-fiber map is consequently the prescribed φ_0 , and its entire enveloping map is $U(\varphi_0)$. One can also check the graded Lie assertion directly from the leading congruences. For $x \in F^d L$ and $z \in F^e L$, write $u = \Phi(x) - f(x)$ and $v = \Phi(z) - f(z)$. Then

$$\Phi([x, z]) - [f(x), f(z)] = [u, \Phi(z)] + [f(x), v] \in (I')^{d+e+1}.$$

The leading congruence for $[x, z] \in F^{d+e} L$, followed by the intersection statement of [theorem 3.2](#), therefore gives

$$f([x, z]) - [f(x), f(z)] \in F^{d+e+1} M.$$

This is bracket preservation on the associated graded algebra, with precisely the symbols supplied by Φ . If another choice of representatives produces a lift $\tilde{f}$, the same intersection statement gives $(f - \tilde{f})(F^d L) \subset F^{d+1} M$. Hence the graded identification is independent of the chosen lift. The individual lifts serve only to choose compatible finite bases for later constructions; they do not replace the marked graded map by a different isomorphism.

Neither the original brackets nor the embedded subalgebra $U(N)$ have been asserted to be preserved by the chosen lift f or by Φ , respectively.

4 Rees families and Laurent bounds

We first record a reduction that can be applied to the original pair, before augmentation normalization or any choice of filtration.

Lemma 4.1. *The given isomorphism $\Phi_{\text{raw}} : U_k(L) \simeq U_k(M)$ is the scalar extension of an isomorphism $\Phi_{K,\text{raw}} : U_K(L_K) \simeq U_K(M_K)$ over a subfield $K \subseteq k$ finitely generated over $\mathbb{Q}$. If L is solvable, so is L_K . The field K embeds in $\mathbb{C}$.*

Proof. Choose finite ordered bases $x_1, \dots, x_a$ and $y_1, \dots, y_b$. Take all coefficients in their two bracket tables and in the finite PBW expansions of $\Phi_{\text{raw}}(x_i)$ and $\Phi_{\text{raw}}^{-1}(y_j)$. If the corresponding supports are S_i and T_j , this uses at most

$$a^3 + b^3 + \sum_i |S_i| + \sum_j |T_j|$$

scalars. Let K be the field they generate over $\mathbb{Q}$. The same bracket tables define Lie algebras over K : their identities are reflected by the inclusion into k . Their scalar extensions recover the original algebras.

The evaluation maps from the smaller enveloping algebras to the original ones are injective by PBW, since they preserve the indexed ordered monomials. The displayed generator images therefore satisfy the defining relations over K , and define algebra maps in both directions. Their inverse identities are likewise reflected by evaluation. This descends the specified Φ_{raw} , including its inverse. Solvability descends through the injective Lie map from L_K to L with scalars restricted. There is no need to descend an infinite list of enveloping elements: the finitely many generator images determine each algebra map. Including the inverse images ensures that the resulting map over K is an equivalence, rather than merely a homomorphism whose scalar extension is the desired forward map.

Finally, embed a finite transcendence basis of $K/\mathbb{Q}$ as an algebraically independent complex family. Extend that embedding across the remaining finite algebraic extension into $\mathbb{C}$. $\square$

We may apply the preceding intrinsic argument over K . A Lie isomorphism obtained there will extend to k ; no embedding of k in $\mathbb{C}$ is needed. We henceforth rename the K -models L, M and their equivalence Φ_{raw} . Normalize this descended map as in [theorem 2.2](#) and again denote it by Φ . The character used in that normalization belongs to K : its values are constant PBW coefficients already included in K . All adapted choices and the marking φ_0 are now made over K .

4.1 The polynomial families

Choose bases $x_1, \dots, x_n$ and $y_1, \dots, y_n$ adapted to the recovered filtrations and matched by the recovered graded isomorphism φ_0 . Write $w_i \geq 0$ for the common weights. With $\mathcal{F}^q U(M)$ denoting the span of PBW monomials of weight at least q , the recovered marking gives

$$\Phi(x_i) - y_i \in \mathcal{F}^{w_i+1} U(M), \quad \Phi^{-1}(y_i) - x_i \in \mathcal{F}^{w_i+1} U(L). \quad (4.1)$$

These follow from [theorem 3.3](#), rather than being additional assumptions on the original isomorphism. The bracket coefficients satisfy $c_{ij}^p \neq 0 \Rightarrow w_p \geq w_i + w_j$, and similarly for M . Distinguish the weight $w(\alpha) = \sum_i w_i \alpha_i$ from the ordinary degree $|\alpha| = \sum_i \alpha_i$. Concretely, choose vector-space sections of the finitely many graded pieces and lift a homogeneous basis through them. Use φ_0 to choose the corresponding target basis. These are linear splittings, with no assertion that either splitting preserves brackets. The filtration is the span of basis vectors of weight at least its index.

Thus $[F^p L, F^q L] \subseteq F^{p+q} L$ gives the displayed support condition. The degree-zero part and the zero-dimensional case are included.

Define the free $K[t]$ -module L_t by

$$[x_i, x_j]_t = \sum_{p: c_{ij}^p \neq 0} c_{ij}^p t^{w_p - w_i - w_j} x_p. \quad (4.2)$$

All exponents are nonnegative. After extending to $K((t))$, the diagonal map $\delta_L(x_i) = t^{-w_i} x_i$ preserves this bracket:

$$\delta_L([x_i, x_j]_t) = \sum_p c_{ij}^p t^{-w_i - w_j} x_p = [\delta_L(x_i), \delta_L(x_j)].$$

Its restriction to the polynomial module is injective. Hence Jacobi is reflected from the original scalar-extended Lie algebra. Construct M_t similarly. At $t = 0$ exactly the equal-weight terms survive, giving the common graded Lie algebra g_0 ; at $t = 1$ the original algebras return.

We use PBW over $K[t]$, and later over $K[t, h]$. For these free Lie modules, ordered monomials span by commutator rewriting. Mapping to the enveloping algebra over the fraction field, field PBW shows that every finite relation between these monomials has zero coefficients. Thus they are a basis over the polynomial ring, and localization is injective. This argument does not presuppose the injectivity it establishes. Symmetrization is triangular with diagonal one for ordinary degree, so it and its inverse preserve the ordinary degree filtration.

Proposition 4.2. *There is a $K[t]$ -algebra isomorphism*

$$\Phi_t : U_{K[t]}(L_t) \simeq U_{K[t]}(M_t)$$

whose zero specialization is the identity under the fixed marking. If $\Phi(x_i) = \sum_{\alpha} a_{i\alpha} y^{\alpha}$, then

$$\Phi_t(x_i) = \sum_{\alpha} a_{i\alpha} t^{w(\alpha) - w_i} y^{\alpha}. \quad (4.3)$$

Proof. Filteredness makes all nonzero terms in (4.3) polynomial. After Laurent scalar extension the formula is $U(\delta_M)^{-1} \Phi U(\delta_L)$, so it preserves the relations. Injective localization reflects these relations to the polynomial envelope. Applying the same construction to Φ^{-1} gives a polynomial inverse; the inverse identities can again be checked after localization. By (4.1), $\Phi_t(x_i) = y_i + tQ_i(t)$ for an actual polynomial element $Q_i(t) \in U_{K[t]}(M_t)$. The zero specialization fixes every marked generator and therefore the whole envelope. $\square$

Let W be the common coordinate vector space and $A_K = S_K(W)$. In symmetric PBW coordinates put

$$\Psi_t = \text{sym}_{M_t}^{-1} \Phi_t \text{sym}_{L_t}, \quad \Psi_t = \sum_{r \geq 0} t^r \Psi_r, \quad \Psi_0 = \text{id}.$$

Here coefficient extraction defines $\Psi_r \in \text{End}_K(A_K)$; each fixed polynomial input has a polynomial output in t . The map intertwines the two transported enveloping products, not necessarily ordinary commutative multiplication.

4.2 Homogenization and the degree estimate

Introduce an independent parameter h , with $U_h(L_t)$ defined by $xy - yx = h[x, y]_t$. This is the envelope of the bracket multiplied by h and has the same PBW basis. Inverting h gives the algebra isomorphism

$$\theta_{h,L} : U_h(L_t)[h^{-1}] \longrightarrow U(L_t)[h, h^{-1}], \quad x_i \longmapsto hx_i.$$

Both sides of its defining relation map to $h^2[x_i, x_j]_t$. Consequently

$$\Phi_{t,h} = \theta_{h,M}^{-1} \Phi_t \theta_{h,L}, \quad \Phi_{t,h}(x_i) = \sum_{\alpha} a_{i\alpha} t^{w(\alpha) - w_i} h^{1-|\alpha|} y^{\alpha}. \quad (4.4)$$

This construction initially concerns finite polynomials. Its scalar extension to $K((h))[t]$ preserves the outer variable t .

Symmetrization of n generators scales by exactly h^n . Thus the block of Ψ_r from $S^n(W)$ to $S^m(W)$ is multiplied under (4.4) by

$$h^{n-m}. \quad (4.5)$$

Indeed, every ordered product in the normalized permutation average has n factors, and hence acquires the same input multiplier. Inverse scaling contributes h^{-m} to its output component. This establishes the exact identity in symmetric coordinates; an estimate in ordered PBW coordinates alone would not identify the operator used below. Choose $D \geq 1$ bounding the ordinary PBW degrees of all terms in the images of basis generators under both Φ and Φ^{-1} . The finite bases and finite PBW supports ensure such a D exists.

Lemma 4.3. *For every $n, r \geq 0$,*

$$\Psi_r(S^n(W)) \subseteq \bigoplus_{m \leq n+r(D-1)} S^m(W).$$

The same bound holds for the coefficients of the inverse comparison.

Proof. Work in the actual algebra $B_t = U_{K[t]}(M_t)$, with its ordinary PBW filtration $B_{t, \leq d}$. This filtration is multiplicative over $K[t]$. The corrections in $\Phi_t(x_i) = y_i + tQ_i(t)$ belong to $B_{t, \leq D}$: subtracting a linear generator and reweighting its coefficients by powers of t cannot increase this degree bound.

Introduce an auxiliary central variable z and consider $y_i + zQ_i(t)$ in $B_t[z]$. In a product of n factors, a coefficient of z^s selects s corrections and has degree at most

$$(n - s) + sD = n + s(D - 1).$$

Apply this estimate to the finite symmetrization average of the input monomial. Inverse target symmetrization is $K[t]$ -linear and degree-nonincreasing, so it preserves this bound.

Finally evaluate $z = t$. The contribution of a z^s coefficient to t^r has residual polynomial degree $r - s$, and therefore $s \leq r$. Its spatial degree is at most $n + r(D - 1)$. This proves the assertion; interchange source and target for the inverse. Throughout, multiplication remains the genuine t -dependent multiplication in B_t . $\square$

The estimate bounds degree *excess* uniformly over every input degree at fixed r . Its dependence on r is essential.

4.3 The completed operator

Extend coefficients along a fixed embedding $K \hookrightarrow \mathbb{C}$, and now write $A = S_{\mathbb{C}}(W_{\mathbb{C}})$ and $C = D - 1$. For any complex vector space X set

$$X((h)) = \left\{ \sum_{d \geq B} h^d x_d : B \in \mathbb{Z}, x_d \in X \right\}, \quad E = \mathbb{C}((h)), \quad R = E[[t]].$$

There is no finite-dimensionality restriction on the coefficient span. For example $\sum_{j \geq 0} h^j x^j$ belongs to $\mathbb{C}[x]((h))$ but not to $E[x]$. Our function space is $\mathcal{A} = A((h))[[t]]$: each t -coefficient has its own Laurent lower bound.

Proposition 4.4. *The finite comparison (4.4) defines an element*

$$T = \text{id} + \sum_{r \geq 1} t^r T_r \in \mathcal{E} := \text{End}_{\mathbb{C}}(A)((h))[[t]], \quad \text{ord}_h T_r \geq -rC,$$

where the bound holds on the whole coefficient endomorphism. It acts invertibly and R -linearly on $\mathcal{A}$, intertwining the two completed homogenized PBW products.

Proof. Define $T_{r,d}$ on $S^n(W_{\mathbb{C}})$ to be the output component of Ψ_r in degree $n-d$, interpreted as zero for $n-d < 0$. This is an endomorphism of A , since each input has finitely many homogeneous components. The uniform-excess lemma gives $T_{r,d} = 0$ whenever $d < -rC$. Hence $T_r = \sum_{d \geq -rC} h^d T_{r,d}$ is a Laurent series of endomorphisms. Formula (4.5) identifies it with the actual homogenized comparison. On an input of degree at most n , its Laurent support is contained in $[-rC, n]$.

For $F = \sum_{d \geq B} h^d F_d$ and $x = \sum_{j \geq b} h^j x_j$, define

$$[h^q](Fx) = \sum_{d=B}^{q-b} F_d(x_{q-d}).$$

This is finite and has output lower bound $B + b$. Outer convolution gives $[t^r](Tx) = \sum_{p=0}^r T_p(x_{r-p})$. If x_s has lower bound b_s , the latter has lower bound $\min_{0 \leq p \leq r} (b_{r-p} - pC)$. These formulas give the asserted scalar-linear action and preserve operator composition. With $G = T - \text{id}$, the inverse $\sum_{s \geq 0} (-G)^s$ exists because G has positive t -order.

The homogenized PBW products have nonnegative h -exponents as whole binary cochains. More explicitly, a multiplication block from $S^a(W) \otimes S^b(W)$ to $S^m(W)$ acquires the factor h^{a+b-m} under homogenization, and the ordinary PBW filtration gives $m \leq a + b$. Thus each coefficient is a binary cochain on all of A , with lower h -bound zero, and the products extend by the same convolution. On finite polynomial inputs, intertwining follows from the algebra isomorphism (4.4). Extracting coefficients gives equality of the binary coefficient maps; convolution then proves it for all inputs in $\mathcal{A}$. $\square$

This is an explicit action of coefficient series, not an identification with all linear or continuous maps on $\mathcal{A}$. The action is faithful: evaluate it on constant inputs $a \in A$, then extract first the t -coefficient and then the h -coefficient to recover each endomorphism applied to a . The same argument detects equality of multilinear coefficient families. It does not require constant inputs to span the completed module algebraically over R . Likewise spatial scaling is only used on finite polynomials: substituting $x \mapsto h^{-1}x$ in $\sum_{j \geq 0} h^j x^{2j}$ would produce an inadmissible series with no Laurent lower bound. The uniform estimate is what constructs the completed comparison. No lower bound uniform in t -degree is needed; $\sum_{r \geq 0} t^r h^{-r} a$ is allowed. The order of completion is therefore

part of the definition. Equivalently, $\mathcal{A}$ is the inverse limit of its finite t -jets, each with coefficients in $A((h))$. It is complete and separated for the t -adic filtration. There is no condition requiring the coefficients of an element of $A((h))$ to lie in a common finite-dimensional subspace of A .

The full Hochschild space of arity q is $C^q(A, A) = \text{Hom}_{\mathbb{C}}(A^{\otimes q}, A)$, including $C^0(A, A) = A$. We use precisely $C^q(A, A)((h))[[t]]$, with cohomological degree $q - 1$. For a cochain lower bound B and input bounds $b_1, \dots, b_q$, the coefficient at h^d is a finite sum over nonnegative indices with total $d - B - \sum_i b_i$; its output lower bound is $B + \sum_i b_i$. These estimates justify evaluation and composition in these spaces, without enlarging them to arbitrary maps on completed functions.

Finally,

$$\log T = \sum_{s \geq 1} \frac{(-1)^{s+1}}{s} (T - \text{id})^s \in t\mathcal{E}.$$

At outer order r , only $s \leq r$ occur, with finitely many positive compositions of r ; their Laurent bounds add to $-rC$. For $1 \leq s \leq r$ there are exactly $\binom{r-1}{s-1}$ such compositions, so this is a finite sum of Laurent series at every outer order. The inverse series obeys the same lower bound, by the identical calculation without the rational logarithm coefficients. Thus the logarithm is a legitimate full Hochschild gauge operator. It is not claimed to be a derivation of one fixed product. The next section applies formality to the resulting associative comparison.

5 From associative equivalence to Poisson equivalence

We now work over $\mathbb{C}$, using the embedding of the coefficient field chosen above. We write h for the deformation parameter and t for the Rees parameter. In this section W denotes the complex scalar extension of the common vector space of the two Rees families. Put $A = S_{\mathbb{C}}(W)$ and write $\pi_L(t), \pi_M(t)$ for their linear Poisson tensors. Their common specialization is denoted by π_0 . For a complex vector space X , our conventions are

$$X((h)) = X[[h]][h^{-1}], \quad E = \mathbb{C}((h)), \quad R = E[[t]], \quad B = A((h)).$$

The order is completion in h , localization, then completion in t . Each t -coefficient has its own lower Laurent bound. We do not require a bound independent of the t -degree. In particular, When $W \neq 0$, B strictly contains $E[W]$; for $W = 0$ both equal E .

Proposition 5.1. *The marked Rees isomorphism of [theorem 4.2](#) induces an R -algebra automorphism Ψ of $B[[t]]$, congruent to the identity modulo t , such that*

$$\Psi(\{f, g\}_{\pi_L(t)}) = \{\Psi(f), \Psi(g)\}_{\pi_M(t)} \quad (f, g \in B[[t]]).$$

The proposition concerns the indicated completed function algebra, not merely finite polynomial inputs or unrelated finite-order approximations. We prove it using affine formality and a complete filtered comparison theorem.

Full cochains and their completed action

Use the shifted differential graded Lie algebras

$$P_{\text{poly}}^p = A \otimes \Lambda^{p+1} W^*, \quad -1 \leq p \leq \dim W - 1, \quad C_{\text{all}}^p = \text{Hom}_{\mathbb{C}}(A^{\otimes(p+1)}, A), \quad p \geq -1.$$

All completions of these graded complexes are taken degreewise. The first has zero differential and the Schouten bracket; the second has the Gerstenhaber bracket and differential $[m, -]$, where m is

ordinary multiplication. Both contain functions in degree -1 and vanish in lower degrees. In degree zero, $C_{\text{all}}^0 = \text{End}_{\mathbb{C}}(A)$ with its commutator bracket. We normalize bivectors so that the first star commutator is h times their Poisson bracket.

The factorial-normalized Hochschild–Kostant–Rosenberg map sends a wedge of q derivations to $1/q!$ times their alternating multilinear product. It is a quasi-isomorphism into the *full* Hochschild complex [3]. In the polynomial case this follows directly from the Koszul resolution of the diagonal. Indeed, the elements $x_i \otimes 1 - 1 \otimes x_i$ in A^e become independent polynomial variables after a change of coordinates, hence their Koszul complex resolves A [7]. Applying $\text{Hom}_{A^e}(-, A)$ gives zero differentials and the polyvector spaces. The alternating comparison with the free bar resolution computes full Hochschild cohomology; its cochain composite with normalized HKR is the identity. The degree signs between the bar differential and $[m, -]$ are absorbed into this comparison.

Kontsevich’s affine formality theorem supplies an L_{∞} morphism $\mathcal{U}$ from polynomial polyvectors to polynomial polydifferential cochains, with first component HKR [5, §§4.6.2, 6.4 of the arXiv version]. We fix the morphism given by his affine graph construction. Its formulas preserve polynomials, and their real coefficients extend to $\mathbb{C}$. Composing with the strict inclusion into C_{all} gives an L_{∞} quasi-isomorphism with the full target just described.

Here are the elementary convergence facts we will use. An element

$$F = \sum_{j \geq a} h^j F_j \in \text{Hom}_{\mathbb{C}}(A^{\otimes q}, A)((h))$$

acts E -multilinearly on B^q . On inputs with lower bounds b_i , the coefficient of h^d sums over $j + j_1 + \dots + j_q = d$ with $j \geq a$ and $j_i \geq b_i$, a finite set. The same argument proves compatibility with compositions and brackets. The action is faithful, by evaluating on constant polynomial inputs. It is not an identification with all E -linear maps on B . After adjoining t , first split a fixed t -degree in finitely many ways, and then apply this Laurent argument. Consequently identities between these completed operators extend from polynomial inputs by coefficientwise expansion, without any assumption that arbitrary linear maps are continuous.

In particular, a Laurent series of unary cochains has a lower bound for the whole operator, rather than a bound chosen separately after seeing its input. If two such operators have bounds a, b , their composition has bound $a + b$. For an outer series $F = \sum_{r \geq 0} t^r F_r$ acting on $f = \sum_{s \geq 0} t^s f_s$, the coefficient of t^n is $\sum_{r+s=n} F_r(f_s)$. Its Laurent lower bound is the minimum of finitely many sums of bounds, hence is finite. This explains why the bounds may vary with r , and why no exchange of the two completions is implicit. The same reasoning applies to every finite arity occurring in the deformation complexes.

Twisting before localization

Set $\alpha_0 = h\pi_0$ and

$$\mu_{\star, 0} = m + \sum_{j \geq 1} \frac{1}{j!} \mathcal{U}_j(\alpha_0^{\otimes j}).$$

The twisted components are

$$\mathcal{U}_n^{\alpha_0}(z_1, \dots, z_n) = \sum_{j \geq 0} \frac{1}{j!} \mathcal{U}_{n+j}(\alpha_0^{\otimes j}, z_1, \dots, z_n).$$

Every additional base input increases h -order by one. These sums therefore converge before localization, and the twisted differentials are $[\alpha_0, -]$ and $[\mu_{\star, 0}, -]$. The twisted identities follow by comparing finite coefficients in the original identities.

Lemma 5.2. *The twisted linear component induces a quasi-isomorphism*

$$(P_{\text{poly}}((h)), [\alpha_0, -]) \longrightarrow (C_{\text{all}}((h)), [\mu_{\star,0}, -]).$$

Proof. First form its cone over $\mathbb{C}[[h]]$. Write C for the underlying graded vector space of the ordinary HKR cone. Its underlying module is $C[[h]]$, with differential $d = d_0 + O(h)$, and its reduction modulo h is the acyclic HKR cone. Choose a contracting homotopy s_0 for this vector-space complex and extend it coefficientwise. Then

$$\mathcal{K} = ds_0 + s_0d = 1 + O(h)$$

is invertible by a geometric series. Since $d^2 = 0$, $\mathcal{K}$ and $\mathcal{K}^{-1}$ commute with d . Thus $s = s_0\mathcal{K}^{-1}$ satisfies $ds + sd = 1$. Every coefficient is finite: $\mathcal{K} - 1$ increases h -order. No polynomial degree bound on s_0 is needed. Localizing this contraction gives a contraction of the Laurent cone, proving the assertion. $\square$

The higher twisted components also extend to Laurent inputs: a fixed finite list of such inputs has a finite sum of lower bounds, while the number of base insertions still increases h -order. Notice that quasi-isomorphism was proved for the completed cone *before* inverting h .

Complete filtered gauge reflection

Over E , form

$$D_P = t(P_{\text{poly}}((h)), [\alpha_0, -])[[t]], \quad D_C = t(C_{\text{all}}((h)), [\mu_{\star,0}, -])[[t]],$$

with $F^r = t^r(-)[[t]]$, $r \geq 1$. These filtrations are separated and complete, start with the whole space, and satisfy $[F^r, F^s] \subset F^{r+s}$. The twisted Taylor maps preserve sums of filtration degrees. At a fixed t -degree r , their MC series uses at most r non-base inputs and finitely many partitions of r ; the remaining Laurent sums are already justified.

On every F^r , the linear component is the product of copies of the map in [theorem 5.2](#), indexed by t -degrees at least r . Its cone is contractible by applying the preceding contraction to each coefficient. Hence it is a quasi-isomorphism on *every* filtration step. All hypotheses of the complete filtered Dolgushev–Rogers theorem are now satisfied over the characteristic-zero field E [[2](#), Theorem 2.2]. We obtain a bijection on connected components of the completed MC simplicial sets. The theorem is applied to these t -positive objects, not directly to all Laurent polyvectors: positivity provides pronilpotence even though negative powers of h occur. The differentials are independent of t , so the coefficientwise contraction really preserves each F^r . The shifted convention used in the cited theorem is obtained by shifting our complexes by one; it imposes no extra nonnegativity condition on our cochain degrees.

For these DGLAs the components are gauge orbits under $\exp(F^1D^0)$. Indeed, an MC one-simplex has the form $\beta(s) + ds\xi(s)$; its equation is the gauge differential equation. Modulo t^r , the time-ordered solution with identity initial value involves finitely many iterated integrals of polynomials in s , defined over any characteristic-zero field. Its uniqueness makes these solutions compatible as r increases, and their limit gives one completed gauge element. A finite chain of one-simplices gives a product of these elements. Conversely a gauge exponential gives an MC path. Degree -1 does not add a component to a one-simplex, because the interval has no nonzero two-forms. Thus we reflect genuine completed gauge equivalence, rather than choose unrelated equivalences in all finite quotients.

For a differential $[a_0, -]$, the gauge action is characterized by

$$a_0 + e^X \cdot \beta = e^{\text{ad } X}(a_0 + \beta).$$

Equivalently the action on β is $e^{\text{ad } X}\beta - \sum_{j \geq 0} (\text{ad } X)^j([a_0, X])/(j+1)!$. We use this with $a_0 = \alpha_0$ and $a_0 = \mu_{\star,0}$, respectively.

The actual associative comparison

Let $\star_L(t), \star_M(t)$ be the star products obtained from the same $\mathcal{U}$. For a linear Poisson tensor, Kontsevich's linear-Poisson comparison gives

$$x \star y - y \star x = h[x, y]_t \quad (x, y \in W),$$

and its top polynomial-degree product is ordinary multiplication [5, §8.3.1 of the arXiv version]. Universality and PBW triangularity therefore give an algebra isomorphism

$$I_L : U_h(L_t) \longrightarrow (A[t, h], \star_L), \quad I_L(x) = x,$$

and similarly I_M .

These are compatible family maps. Their values on monomials are products of the corresponding linear generators. A graph term of order h^j for a linear Poisson tensor contributes j linear coefficients and $2j$ derivatives, so it lowers total polynomial degree by j . It vanishes when that resulting degree is negative. Thus each polynomial calculation is finite and polynomial in the structure constants, and works over $\mathbb{C}[t, h]$. Define

$$J_L = I_L \circ \text{sym}_L, \quad J_M = I_M \circ \text{sym}_M.$$

They intertwine symmetric PBW products with star products, fix linear functions, and have the form $1 + O(h)$. Their inverses exist by the h -adic geometric series, with no negative h -powers. Consequently all four operators belong to $\text{End}_{\mathbb{C}}(A)[[h]][[t]]$. Since the two brackets agree at zero, the same construction gives $J_L(0) = J_M(0) = J_0$.

More explicitly, $J_L(x_{i_1} \cdots x_{i_q})$ is the average over permutations of the q corresponding star factors. Thus its value at the common special bracket is forced; no pointwise choice of an isomorphism remains. Triangularity proves bijectivity by induction on polynomial degree, and the same induction works with polynomial t -coefficients. Passing to series then uses the already defined coefficientwise action. Ordinary PBW symmetrization is only a linear identification here. It is J_L , and respectively J_M , that converts its transported product into the chosen star product.

Write T for the completed homogenized Rees comparison supplied by [theorem 4.4](#). Its essential conclusion is

$$T \in 1 + t \text{End}_{\mathbb{C}}(A)((h))[[t]], \quad \text{ord}_h(T_r) \geq -r(D-1),$$

with the bound valid simultaneously on all polynomial inputs. Here T_r is the coefficient of t^r , and D is the finite ordinary PBW degree bound from the preceding section. In symmetric coordinates a block from degree a to degree b is multiplied by h^{a-b} ; the bound $b \leq a + r(D-1)$ is precisely what gives the displayed estimate uniformly over all a . This identification retains the actual original enveloping map, rather than introducing an abstract equivalence between deformation classes. It intertwines the actual completed PBW products. The identity on finite polynomial inputs extends to $B[[t]]$ by the finite-convolution argument above; no globally defined inverse degree-scaling map on B is asserted.

Now set

$$G = J_M T J_L^{-1}.$$

This is an isomorphism from $\star_L$ to $\star_M$ on $B[[t]]$, and $G(0) = J_0 \text{id } J_0^{-1} = \text{id}$. For each t -degree, its coefficients are finite sums of compositions of uniformly bounded Laurent operators with nonnegative h -series. Therefore

$$X = \log G \in t \text{End}_{\mathbb{C}}(A)((h))[[t]] = D_C^0.$$

At outer degree r , the logarithm uses at most r positive factors; this proves its existence in the stated operator algebra. Its coefficients need not be polydifferential, which is why the full Hochschild target was necessary.

If m_L, m_M denote the completed star products, multiplicativity says

$$m_M = G \circ m_L \circ (G^{-1} \otimes G^{-1}) = e^{\text{ad } X} m_L.$$

The second identity follows by differentiating conjugation by e^{sX} ; its infinitesimal action is the Gerstenhaber bracket. This supplies a target gauge equivalence relative to the common $\mu_{\star,0}$.

Proof of [theorem 5.1](#). Set $\beta_L = h(\pi_L(t) - \pi_0)$ and $\beta_M = h(\pi_M(t) - \pi_0)$. They are MC elements of D_P . Expansion of the twisted Taylor sums gives exactly

$$(\mathcal{U}^{\alpha_0})_*(\beta_L) = m_L - \mu_{\star,0}, \quad (\mathcal{U}^{\alpha_0})_*(\beta_M) = m_M - \mu_{\star,0}.$$

Grouping terms by their base and positive- t inputs produces the factors $1/j!$ and $1/n!$; all rearrangements are justified by the two coefficientwise convergence arguments. Thus the constructed G compares precisely these two MC images. Injectivity on connected components reflects this equivalence to the source.

A source gauge element is the exponential of a t -positive vector field with coefficients in $B[[t]]$. Its action on functions is well-defined by Laurent convolution and t -adic convergence. The coefficientwise Leibniz rule implies that its exponential is an R -algebra automorphism, with inverse the opposite exponential, and it is the identity modulo t . The source gauge equation gives the required Poisson intertwining for $h\pi_L(t)$ and $h\pi_M(t)$, with the direction reversed if necessary by inversion. Cancel the invertible scalar h to obtain the assertion. $\square$

6 The first jet and marked descent

Let $K \subseteq k$ be the coefficient field supplied by [theorem 4.1](#), with its chosen embedding into $\mathbb{C}$. Write L_K, M_K for the resulting Lie algebras. The marking of [theorem 3.3](#) is defined over K :

$$\varphi_0 : \text{gr}_F L_K \longrightarrow \text{gr}_F M_K.$$

Choose adapted bases whose graded classes correspond under φ_0 , and write w_i for their common weights. Thus F^a is spanned by the vectors of weight at least a , and the marking is the identity in these graded bases. Fix c with $F^{c+1} = 0$. We retain the nilradical filtration, including $F^0 g = g$, throughout this section.

6.1 Extracting the first jet in the actual completed algebra

Put

$$A = \mathbb{C}[x_1, \dots, x_n], \quad B = A((h)), \quad E = \mathbb{C}((h)), \quad R = E[[t]].$$

An element of $B[[t]]$ has the form

$$\sum_{r \geq 0} t^r \sum_{q \geq q_r} h^q f_{rq}(x), \quad f_{rq} \in A.$$

The lower bound q_r may depend on r , and the coordinate degrees of f_{rq} need not be bounded. For $n > 0$, this function algebra strictly contains $E[x_1, \dots, x_n][[t]]$: the inner series $\sum_{q \geq 0} h^q x_1^q$ belongs to B but has unbounded coordinate degree. For $n = 0$, both algebras are simply R . The argument below extracts coefficients of coordinate degree zero and one before making any finite-dimensional construction; it never assumes polynomial dependence on the coordinates over E .

A t -small vector field with coefficients in $B[[t]]$ acts as an R -linear derivation of this algebra. Coordinate differentiation preserves the lower Laurent bound; multiplication by any fixed coefficient shifts it by a finite amount. At fixed outer order, composition involves only finitely many outer indices. Its exponential is therefore defined t -adically: at order t^r , only its first r powers contribute. The iterated derivation rule proves multiplicativity, and exponentiating the negative vector field gives the inverse. These observations justify the action on the indicated completed algebra, without requiring an action on a larger coordinate-adic completion.

By [theorem 5.1](#), we have an R -algebra isomorphism $\Psi : B[[t]] \rightarrow B[[t]]$, congruent to the identity modulo t , comparing the linear Poisson structures of the two Rees families. A common factor h in these structures can be cancelled because $h \in R^\times$. We orient the comparison from source coordinate functions to target coordinate functions.

Lemma 6.1. *The matrix*

$$P_{ai} = \partial_a \Psi(x_i)|_{x=0}$$

is defined over R , belongs to $\mathrm{GL}_n(R)$, specializes to the identity, and is a Lie isomorphism between the scalar-extended Rees families. No preservation of augmentation by Ψ is required.

Proof. Define augmentation $\epsilon : B[[t]] \rightarrow R$ and coordinate derivatives coefficientwise in both parameters. Evaluation of each polynomial at zero, and its coordinate derivatives, preserve its Laurent lower bound. Consequently ϵ is an algebra homomorphism and ∂_a is an R -linear derivation, with

$$\epsilon(x_a) = 0, \quad \epsilon(\partial_a x_b) = \delta_{ab}.$$

In particular $P_{ai} = \epsilon(\partial_a \Psi(x_i))$ is well-defined even with unbounded coordinate degree. The marking gives $P \equiv I \pmod{t}$. Writing $P = I + tS$, the geometric series $\sum_{j \geq 0} (-tS)^j$ proves invertibility over R itself.

Let $C_{ij}^{L,k}(t)$ and $C_{ab}^{M,r}(t)$ denote the two Rees structure tables. For arbitrary completed functions,

$$\{f, g\}_M = \sum_{a,b,r} C_{ab}^{M,r}(t) x_r (\partial_a f) (\partial_b g).$$

Its constant term vanishes. Moreover, the product rule gives

$$\epsilon \partial_s \{f, g\}_M = \sum_{a,b} C_{ab}^{M,s}(t) \epsilon(\partial_a f) \epsilon(\partial_b g).$$

Every term containing a second derivative still has the factor x_r and vanishes under ϵ . This finite identity establishes the first-jet calculation directly in the completed algebra.

Apply it to the Poisson identities

$$\{\Psi(x_i), \Psi(x_j)\}_M = \sum_k C_{ij}^{L,k}(t) \Psi(x_k).$$

If $c_i = \epsilon(\Psi(x_i)) \in tR$, their constant terms give $\sum_k C_{ij}^{L,k}(t) c_k = 0$: the constant part defines a character of the source. Their first derivatives give

$$\sum_{a,b} P_{ai} P_{bj} C_{ab}^{M,s}(t) = \sum_k C_{ij}^{L,k}(t) P_{sk}. \quad (6.1)$$

These are exactly the Lie-matrix equations. There is no need to translate the character to zero. Nor do we claim that Ψ induces a map on the augmentation ideal modulo its square: that ideal need not be preserved. Invertibility came from the marking, rather than such a claim. $\square$

6.2 A regular arc with the prescribed graded special point

Let $E_t = E((t))$. With our decreasing-filtration convention,

$$[x_i, x_j]_{L_t} = \sum_r c_{ij}^{L,r} t^{w_r - w_i - w_j} x_r, \quad \delta_{L,t}(x_i) = t^{-w_i} x_i.$$

The latter map identifies the generic Rees algebra with $L_K \otimes_K E_t$. The corresponding target scaling gives

$$f_{E_t} = \delta_{M,t} P \delta_{L,t}^{-1} : L_K \otimes_K E_t \xrightarrow{\sim} M_K \otimes_K E_t, \quad (f_{E_t})_{ri} = t^{w_i - w_r} P_{ri}.$$

Every isomorphism of the original Lie algebras preserves the nilradical and its lower central series. Their previously established compatibility with scalar extension therefore makes f_{E_t} filtered. Hence its entries with $w_r < w_i$ are zero. Since $R \hookrightarrow E_t$ is injective and t is invertible in E_t , the same entries of P are zero. More explicitly, the column of weight w_i belongs to F^{w_i} , whose adapted basis has no vector of smaller weight. This makes the corresponding coefficient of f_{E_t} exactly zero, rather than merely a series divisible by a power of t . Filtration preservation of P has thus been deduced, not imposed on the Poisson comparison.

On equal-weight blocks the scaling factors cancel. The block diagonal matrix

$$y(t) = \bigoplus_a P_{aa}(t)$$

is a regular graded isomorphism over R , with $y(0) = \varphi_0$. Here a indexes weights, rather than individual basis vectors. Each block is congruent to the identity modulo t , so its inverse and the inverse of the full determinant are regular. The graded bracket equations hold in E_t and hence in R . The off-diagonal entries of f_{E_t} may have poles; they are never specialized at zero.

Proposition 6.2. *The marked Rees comparison above yields an isomorphism $L_K \simeq M_K$ over K whose associated-graded map is the prescribed φ_0 .*

We first obtain a marked point over a finite Galois extension, then prove that its descent obstruction vanishes.

The filtered isomorphism scheme $\mathcal{I} = \text{Isom}_F(L_{\mathbb{C}}, M_{\mathbb{C}})$ is affine of finite type. In adapted bases it is defined by the bracket equations, the forbidden-entry equations $f_{ri} = 0$ for $w_r < w_i$, and $z \det f = 1$ with one extra coordinate z . An invertible block triangular matrix has invertible diagonal

blocks over every commutative coefficient ring, since their determinant product is a unit. Thus taking diagonal blocks defines a regular map on $\mathcal{I}$ and a homomorphism

$$\rho : \mathcal{A} = \text{Aut}_F(M_{\mathbb{C}}) \longrightarrow \text{Aut}_{\text{gr}}(\text{gr}_F M_{\mathbb{C}}).$$

The inverse determinant of the diagonal matrix equals z , so this morphism is polynomial in the chosen affine coordinates.

The point f_{E_t} makes $\mathcal{I}$ nonempty. The Nullstellensatz supplies a complex point f_* ; see [8]. Composition with f_* identifies $\mathcal{I}$ with $\mathcal{A}$. The algebraic-group homomorphism theorem factors ρ through a closed subgroup H , with $\mathcal{A} \rightarrow H$ faithfully flat. The smooth-group version suffices in characteristic zero by [6, Theorems 1.72 and 3.23]. Each complex point of H lifts, because its fiber is nonempty and of finite type over an algebraically closed field. Consequently the graded image of $\mathcal{I}(\mathbb{C})$ is the closed coset

$$Y = H \text{gr}(f_*) \subseteq \text{Isom}_{\text{gr}}(\text{gr}_F L_{\mathbb{C}}, \text{gr}_F M_{\mathbb{C}}).$$

This does not assert surjectivity on rational points over K . In particular, one cannot replace the following specialization argument by the assertion that a generic isomorphism has a limit as an ordinary isomorphism: its off-diagonal entries need not have such a limit. Closedness is applied only after passing to the graded image, where the entire arc, including the inverse determinant coordinate, is regular.

For completeness, the generic-point implication can be checked entirely in finite polynomial coordinates. Let I define $\mathcal{I}$, and let d be its diagonal-coordinate map. If a polynomial q vanishes on $Y(\mathbb{C})$, then $q \circ d$ vanishes on all complex solutions of I . The Nullstellensatz gives $(q \circ d)^m \in I$ for some $m \geq 1$. Evaluation at f_{E_t} implies $q(d(f_{E_t}))^m = 0$, hence $q(d(f_{E_t})) = 0$ in the field E_t . This verifies extension-field containment without assuming that the defining ideal is radical.

It follows that $q(y(t)) \in R$ becomes zero in E_t , so it is already zero in R . Specializing gives $q(\varphi_0) = 0$ in E . Since φ_0 and q have complex coefficients, injectivity of $\mathbb{C} \rightarrow E$ gives the same equality over $\mathbb{C}$. Thus $\varphi_0 \in Y(\mathbb{C})$ and a complex filtered isomorphism with precisely this mark exists.

The scheme $X = \text{Isom}_{F, \varphi_0}(L_K, M_K)$ is defined by finitely many equations over K and has a complex point. Its coordinate algebra is nonzero. A maximal ideal has residue field finite over K by [8]; enlarging that field to its normal closure gives a finite Galois extension K_1/K and a point $f \in X(K_1)$. No algebraicity of E or E_t over K was needed.

6.3 The unipotent kernel and finite Galois averaging

Let

$$U = \ker(\text{Aut}_F(M_K) \longrightarrow \text{Aut}_{\text{gr}}(\text{gr}_F M_K)).$$

For any commutative K -algebra S , an element $u \in U(S)$ satisfies $(u - I)F^i \subseteq F^{i+1}$ for every $i \geq 0$, and therefore $(u - I)^{c+1} = 0$. The condition includes the weight-zero quotient $M_K/N(M_K)$. Thus U is a closed subgroup of a unitriangular group, with no general linear factor left on that quotient.

Let $\mathcal{A}_r(S)$ be the endomorphisms raising the filtration by at least r , let $U_r(S) = \{u \in U(S) : u - I \in \mathcal{A}_r(S)\}$, and let $D_r(S) = \text{Der}_S(M_S) \cap \mathcal{A}_r(S)$. For $r \geq 1$ these satisfy

$$U_1 = U, \quad U_{c+1} = 1, \quad \mathcal{A}_r \mathcal{A}_s \subseteq \mathcal{A}_{r+s}.$$

The spaces D_r are defined by linear equations over K , so they commute with scalar extension. Finite polynomial formulas give

$$\exp D = \sum_{j=0}^c \frac{D^j}{j!}, \quad \log u = \sum_{j=1}^c \frac{(-1)^{j+1}}{j} (u - I)^j. \quad (6.2)$$

We justify, in particular, that the logarithm is a derivation. Put $T = u - I$ and $u^s = \sum_{j=0}^c \binom{s}{j} T^j$. At nonnegative integers s this is an ordinary power of the Lie automorphism u , so $u^s[a, b] = [u^s a, u^s b]$. These bounded-degree polynomials agree identically: evaluation at sufficiently many integers has a Vandermonde determinant which is a nonzero integer, hence a unit in every S . Differentiating at zero gives the derivation rule for $\log u$, since $\binom{s}{j}'|_0 = (-1)^{j-1}/j$.

Conversely, the identity $D^m[a, b] = \sum_j \binom{m}{j} [D^j a, D^{m-j} b]$ proves that $\exp D$ preserves brackets. Exponential and logarithm are inverse by the formal one-variable identities evaluated on nilpotent operators. All formulas are finite and natural in S . Thus U is isomorphic as a scheme to the finite-dimensional affine space D_1 ; it is smooth and connected. Its group law need not become addition under this isomorphism. The common finite filtration is essential here: arbitrary individually nilpotent endomorphisms would not ensure that products of sufficiently many different operators vanish. Here every such product raises the same filtration beyond its last nonzero term. The inverse of $\exp D$ is $\exp(-D)$, which also preserves the filtration and has identity graded map. These facts establish that all corrections used below are actual Lie automorphisms, not merely invertible linear maps.

Lemma 6.3. *For this group U over any characteristic-zero field K , every finite Galois cocycle is a coboundary. In particular $H^1(K, U) = 1$.*

Proof. Let $\Gamma = \text{Gal}(K_1/K)$ and $a_{\sigma\tau} = a_\sigma \sigma(a_\tau)$ with $a_\sigma \in U(K_1)$. Assume inductively that its values lie in $U_r(K_1)$. In the vector space $\mathcal{A}_r/\mathcal{A}_{r+1}$ put

$$A_\sigma = [a_\sigma - I] = [\log a_\sigma].$$

The equality holds because $2r \geq r + 1$. Products of two differences vanish in this quotient, so $A_{\sigma\tau} = A_\sigma + \sigma A_\tau$. Define

$$D = \frac{1}{|\Gamma|} \sum_{\tau \in \Gamma} \log a_\tau \in D_r(K_1), \quad v = [D].$$

Averaging gives $\sigma v = v - A_\sigma$. For $b_r = \exp(-D)$ the transformed cocycle $a'_\sigma = b_r a_\sigma \sigma(b_r)^{-1}$ has first-order class $-v + A_\sigma + \sigma v = 0$. Hence it takes values in U_{r+1} . The transformed elements still satisfy the cocycle identity: in the product $a'_\sigma \sigma(a'_\tau)$ the adjacent factors $\sigma(b_r)^{-1} \sigma(b_r)$ cancel. Galois acts on the filtration and the defining derivation equations because both are defined over K . Thus the averaging and correction can be repeated with the same action. After the finitely many stages $r = 1, \dots, c$, the transformed cocycle is one. With $b = b_c \cdots b_1$, then

$$b a_\sigma \sigma(b)^{-1} = 1, \quad a_\sigma = b^{-1} \sigma(b).$$

Every sum is finite and its denominator is invertible in characteristic zero. Finally, a torsor under this smooth finite-type group acquires a point over an algebraic closure, whose finitely many coordinates belong to a finite Galois extension. Thus the finite assertion proves the stated Galois-cohomological conclusion. $\square$

Completion of the proof of theorem 6.2. The group U acts on X by postcomposition. For two marked points f_1, f_2 , their difference $f_2 f_1^{-1}$ lies in U . These formulas are regular and functorial, so the geometrically nonempty X is a U -torsor. For its point $f \in X(K_1)$ set $a_\sigma = f(\sigma f)^{-1}$. The mark is defined over K , and therefore $a_\sigma \in U(K_1)$. Direct multiplication gives

$$a_\sigma \sigma(a_\tau) = f(\sigma f)^{-1} (\sigma f)(\sigma \tau f)^{-1} = a_{\sigma\tau}.$$

By [theorem 6.3](#), write $a_\sigma = b^{-1}\sigma(b)$. Since $\sigma f = a_\sigma^{-1}f$, we have $\sigma(bf) = \sigma(b)a_\sigma^{-1}f = bf$. All entries of bf lie in K . Its determinant is nonzero, its bracket and filtration equations hold over K , and its graded map is φ_0 because $b \in U$. This is the required marked isomorphism. $\square$

Proof of [theorem 1.1](#). Descend the given associative equivalence, including its inverse, to the coefficient field $K \subseteq k$ using [theorem 4.1](#). Solvability descends to the source model. Apply the intrinsic filtration argument over K , obtaining the actual marking of [theorem 3.3](#). Embed K into $\mathbb{C}$, construct the compatible Rees families, and use [theorems 5.1](#) and [6.1](#) to obtain their identity-marked Lie matrix over $\mathbb{C}((h))[[t]]$. [Theorem 6.2](#) gives a Lie isomorphism over K . Extending scalars along $K \subseteq k$ gives $L \simeq M$ over k . Thus the argument applies to every characteristic-zero ground field; no embedding of that entire field into $\mathbb{C}$ is assumed. $\square$

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